



UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
WASHINGTON, D.C. 20460

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OFFICE OF
RESEARCH AND DEVELOPMENT

MEMORANDUM

SUBJECT: Cadillac Regional Area Risk Assessment Review
Northernnaire Site, Cadillac, Michigan

FROM: Malcolm S. Field *Malcolm S. Field*
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Exposure Assessment Group (RD-689)

TO: Bruce Means
Office of Solid Waste and
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THRU: Seong T. Hwang, Acting Branch Chief *S. T. Hwang*
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This risk assessment document was prepared for Region V to "...estimate the magnitude and probability of actual or potential harm to public health or the environment from exposure to chemicals in the groundwater in the absence of remediation". However, the material forwarded to this office (Exposure Assessment Group) did not include the first seven chapters of the risk assessment document. A telephone call to the regional contact person was made in an attempt to obtain the first seven chapter and additional material describing the site hydrogeology (remedial investigations and supplements, pumps test analysis, and the record of decision). These were immediately mailed to this office by the region, but have become lost in transit. As a result of this material being lost in the mail, a review of the risk assessment is necessarily constrained to the available information.

The selection of chemicals of concern resulted in four chemicals being eliminated from further consideration:

1,1-dichloroethane (low frequency of detection);
ethylbenzene (low frequency of detection);
xylenes (low frequency of detection); and
benzene (low frequency and low concentration)!



These may be appropriate, but certain factors should be kept in mind. Chemicals like 1,1-dichloroethane and to a lesser extent, benzene, have a low partitioning coefficient, are soluble in water and do not possess a large specific gravity. In addition 1,1-dichloroethane is not well represented by 1,1,1-trichloroethane as is supposed. 1,1,1-trichloroethane is much denser, less soluble in water, has a higher partitioning coefficient, and is soluble in several different organics. Ethylbenzene and xylenes are said to be adequately represented by toluene, but discrepancies do exist, mostly in their respective solubilities and partitioning coefficients.

Table I provides a listing of the more relevant parameters to keep in mind when evaluating plumes. The important aspects to bear in mind are that each chemical is going to partition between the solid, liquid, and gas phases in different manners, and several are soluble with each other. Without actually knowing the soil types in this area, it is difficult to draw very specific conclusions about how each chemical will persist in the subsurface. However, in general, certain aspects may be estimated.

1. specific gravities greater than one are likely to sink;
2. specific gravities less than one are likely to float;
3. large water solubilities are likely to result in facilitated transport by ground water;
4. solubilities in organics result in the probability of cosolvation;
5. high vapor pressures will allow for volatilization in the subsurface;
6. low partitioning coefficients allow for subsurface migration as a liquid;
7. high partitioning coefficients result in the adsorption onto organic carbon in the soil--and little migration; and
8. moderate partitioning coefficients result in an eventual equilibrium being established between adsorption and desorption.

Taking these all together is very difficult to visualize and to model. Yet it is this set of factors coupled together and with ground water specifics that must be accurately modeled to determine the real threat posed. Problems such as, cosolvation, may indicate that chemicals with a higher partitioning

coefficient (ethylbenzene) may actually be easily dissolved and freely transported.

The models used to evaluate the problems at this site are analytical expressions because the "...availability of data and the current state of the art for contaminant transport modeling rarely support the use of sophisticated transport models". This statement is not necessarily correct. Analytical expressions are used for simplified physical conditions that represent the behavior of particular physical processes when isolated from other effects.

The need to gain high levels of resolution in the prediction of contaminant migration and concentration generally requires a numeric model that is capable of simulating site-specific geometry and dimensionality.

All models have certain limiting factors commonly known as assumptions. The more complex the model, the less limiting the assumptions are. The solution chosen for this site requires:

1. uniform one-dimensional flow;
2. homogenous, isotropic aquifer;
3. dispersion coefficients are constant; and
4. solute is conservative.

These assumptions are not very realistic. Few aquifers are really homogenous and isotropic or consist of a one-dimensional flow component. The use of a numeric model would allow for variations at selected points to more realistically simulate subsurface transport.

The fact that the analytical solution predicts high risks in the next several years is significant, but the possibility that facilitated transport via cosolvation or just by discrete high permeability zones may warrant the need for a more complex model to ensure that the contamination is not actually moving faster than predicted.

Other factors of concern in this study are such things as:

1. no environmental affects predicted;
2. no surface water discharge locations considered; and
3. contamination possibility of the aquitard.

Without site specific geologic and hydrogeologic information, it is difficult to conduct a meaningful review. So the comments contained herein may or may not be appropriate depending on the site conditions. The one thing that definitely must be dealt with however, is that the contaminate transport simulation should be as accurate as possible. Too many estimated parameters can result in errors that are orders of magnitude off.

Attachment

cc: Mary Elain Gustafson, Region V

TABLE 1
SOME CHEMICAL AND PHYSICAL PARAMETERS OF
CONTAMINANTS DETECTED AT THE CADILLAC AREA

CHEMICAL	SPECIFIC GRAVITY	SOLUBILITY IN WATER	SOLUBILITY IN ORGANICS	VAPOR PRESSURE	PARTITIONING COEFFICIENT (Log K_{ow})
1,1,1-Trichloroethane	1.34	480-4400 mg/l	Acetone, Benzene, Methanol, Carbon Tetrachloride, Ether, Alcohol, & Chlorinated Solvents	123 mmHg	2.17
1,1-Dichloroethane	1.1776	5 g/l	Miscible in Alcohol	180 mmHg	1.79
Acetone	0.7899	Miscible	Soluble in Alcohol, Ether, Benzene & Chloroform	190 mmHg	-0.24
Ethylbenzene	0.867	161 mg/l	Freely soluble in organic solvents	7 mmHg	3.15
Toluene	0.8669	534.8 mg/l	Soluble in Acetone, Ligroin & Carbon Disulfide; Miscible with Alcohol, Ether, Benzene, Chloroform, Glacial Acetic Acid, and other organic solvents	28.7 mmHg	2.69
Trans-1,2-Dichloroethene	1.2565	600 mg/l	Miscible with Alcohol, Ether, & Acetone; very soluble in Benzene & Chloroform	200 mmHg	1.48
Xylenes	0.86	160 mg/l	Soluble in Alcohol, Ether & other organic solvents	10 mmHg	3.00

CHEMICAL	SPECIFIC GRAVITY	SOLUBILITY IN WATER	SOLUBILITY IN ORGANICS	VAPOR PRESSURE	PARTITIONING COEFFICIENT
Benzene	0.879	1780 mg/l	Miscible with Ethanol, Ether, Acetic Acid, Acetone, Chloroform, Carbon Disulfide, Carbon Tetrachloride	75 mmHg	1.95-2.13
Chloroform	1.4832	8200 mg/l	Soluble in Acetone; Miscible with Alcohol, Ether, Benzene, & Ligroin	150.5 mmHg	1.97
1,1-Dichloroethylene	1.218	400 mg/l	Sparingly soluble in Alcohol, Ether, Acetone, Benzene, & Chloroform	500 mmHg	1.48
1,2-Dichloroethane	1.253	8 g/l	Miscible with Alcohol, Chloroform, & Ether	61 mmHg	1.48
Methylene Chloride	1.3266	13,200-20,000 mg/l	Miscible with Alcohol & Ether	362.4 mmHg	1.25
Tetrachloroethylene	1.63	150-200 mg/l	Soluble in Alcohol & Ether, & Benzene	14 mm Hg	2.88
Trichloroethylene	1.4642	1000 mg/l	Soluble in Alcohol, Ether, Acetone & Chloroform	60 mm Hg	2.29
Chromium (Hexavalent)	7.20	Insoluble (Some Compounds Soluble)			
Cyanide	0.699	Soluble (Forms HCN)	Soluble in Alcohol & Ether		